

Class 5 - Mass transfer and Python modeling practice Modelling 2025

Aim for today is to make sure you are comfortable with:

1. Basic heat and mass transfer modes and mechanisms
2. Formulation of lumped parameter heat and mass transfer models
3. Developing simple analytical solutions
4. Interpreting simple analytical solutions
5. Developing more complex numerical solutions
6. The finite difference method (Euler's and `solve_ivp()`)
7. Implementing models in Python
8. User-defined Python functions
9. Well-organized modular approach to Python models

Plan:

- 12.15-12.20: Repo resources
- 12.20-12.25: Menti poll on topics
- 12.25-12.40: Gills solution
- 12.40-12.50: Interphase mass transfer
- 12.50-13.00: Break
- 13.00-14.00: Aeration exercise

Repo resources: docs

See the "docs" directory for some helpful documents

 main ▾ [course-modelling-2025](#) / docs / 

 **sashahafner** tweaks to exercises and more for class tomorrow

Name
 ..
 constitutive_eqs.pdf
 constitutive_eqs.xopp
 exp_decay_growth.pdf
 exp_decay_growth.xopp
 finite difference formulas.md
 github_intro.md
 links.md
 markdown_intro.md
 py_data.md
 py_data.qmd
 py_mutable_reference.md
 py_mutable_reference.qmd
 requirements_file.md
 symbols.md
 terms.md
 warm_up.md

Repo resources: exercise solutions

course-modelling-2025/classes

github.com/AU-BCE-EE/course-modelling-2025/tree/main/classes/03_02_sept

AU-BCE-EE / course-modelling-2025

<> Code

Issues

Pull requests

Actions

Projects

Security

Ins

main

course-modelling-2025 / classes / 03_02_sept / solutions /

sashahafner

add/update gills and coffee solutions

Name	La
..	
coffee_function	ai
coffee_simple	ai
coffee.pdf	ai
goats.pdf	ai
quizzes.pdf	st
tank.pdf	ai
tank.png	ai
tank.py	ai